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tuition

Volume & 3D Shapes

Year 5 Accelerated — Homework ⏱ 45 min

Worked example — read this through before you start

Worked example: What is the volume of a cuboid measuring 6cm × 4cm × 3cm?
Method & answer: Volume = 6 × 4 × 3 = 72 cm³.

Easy / Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. What is the volume of a cuboid measuring 4cm × 5cm × 2cm?

2. What is the volume of a cube with sides of 6cm?

3. Which of these has a volume of 42cm³? A) 7×3×2 B) 6×7×1 C) both A and B D) neither

4. How many edges does a cuboid have?

5. How many vertices does a cuboid have?

6. What is the volume of a cuboid measuring 7cm × 2cm × 3cm?

7. Which 3D shape comes to a single point (apex) with a circular base? A) cylinder B) cone C) sphere D) cube

8. How many faces does a cone have (flat and curved together)?

9. Select ALL of these that are true properties of a cylinder: A) 3 faces B) 2 flat circular faces C) 1 curved surface D) 12 edges E) 8 vertices

10. What is the volume of a cube with sides of 7cm?

11. How many edges does a square-based pyramid have?

12. What is the volume of a cuboid measuring 9cm × 3cm × 2cm?

13. How many vertices does a triangular prism have?

14. How many edges does a cylinder have?

15. What is the volume of a cube with sides of 10cm?

Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. A cuboid has a volume of 84cm³. Its length is 6cm and width is 2cm. What is its height?

2. A triangular prism has 5 faces, 9 edges and 6 vertices. What is the total if you add all three?

3. Convert 4500cm³ to litres.

6. How many vertices does a cone have?

7. A fish tank measures 40cm × 25cm × 20cm. What is its volume in litres?

8. Select ALL of these statements that are TRUE about the net '2 triangles and 3 rectangles': A) it folds to make a triangular prism B) it has 5 faces C) it folds to make a cube D) 2 of its faces are triangles E) it has 6 faces

11. Cuboid A measures 5×2×2cm; cuboid B measures 4×2×3cm. Which has the greater volume?

12. A cube's 6 faces are identical squares, each with an area of 25cm². What is the length of one edge?

13. Convert 2.2 litres to cm³.

4. A cube has a volume of 216cm³. What is the length of one side? A) 6cm B) 12cm C) 36cm D) 216cm

5. A cuboid measures 12cm × 4cm × 3cm. What is its volume?

9. A cuboid's volume is 120cm³. Its length is 6cm and height is 5cm. What is its width?

10. How many faces does a cube have?

14. A cuboid has dimensions 8cm × 3cm × 4cm. What is its volume?

15. How many edges does a triangular prism have?

Standard / Advanced — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. A cube's edge is 2cm longer than 4cm. What is the cube's volume?

2. A cuboid's height is 4 times its width of 2cm, and its length is 5cm. What is its volume? A) 40cm³ B) 80cm³ C) 160cm³ D) 20cm³

3. The volume of a cube is the same as a cuboid measuring 9cm×6cm×4cm. What is the length of the cube's edge?

4. A cuboid's width is a quarter of its length of 20cm, and its height equals its width. Select ALL of these that are TRUE: A) the width is 5cm B) the height is 5cm C) the volume is 500cm³ D) the volume is 400cm³ E) the width is 4cm

5. A cuboid box has a volume of 210cm³. Its length is 7cm and width is 5cm. What is its height?

6. A cuboid measures 6cm × 5cm × ? cm and has a volume of 210cm³. What is the missing length, and the area of its largest face?

7. Convert 1.331 litres to cm³, then find the side length of a cube with that volume.

8. A cuboid's three different face areas are 8cm², 6cm² and 12cm². What is its volume?

9. A 9cm × 10cm × 10cm tank is exactly two-thirds full of water. How many millilitres does it contain? (1cm³ = 1 ml)

10. A tank measures 45cm × 30cm × 25cm and is filled to 80% of its capacity. How many litres of water does it contain?

11. A cube-shaped gift box has a volume of 729cm³. Wrapping paper covers all 6 faces exactly. What total area of wrapping paper is needed?

12. A water butt measures 60cm × 40cm × 80cm and is filled at 8 litres per minute. How long does it take to fill completely?

13. A 24cm × 18cm × 12cm box is packed with 6cm cubes, with no gaps. How many blocks are needed to fill it completely?

14. A tank is 1.2m long, 0.8m wide and 0.5m deep. What is its volume in m³, and how many litres is this?

15. Two identical cube-shaped tanks each hold 64,000cm³. Water is poured from a full first tank into the empty second tank until both contain the same amount. How many litres are then in each tank?